Term Project: First Draft

Short-Rate Modeling, Term Structure, and Interest Rate Derivatives
MATH 60646A – Stochastic Calculus, HEC Montréal, Winter 2026

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A Note on How to Read This Document

This document is written the way an instructor would write notes: every mathematical object is introduced in plain language before any formula appears, every formula is translated back into plain language after it is written, and every key idea is illustrated with at least one concrete example. We work throughout on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, where $\mathbb{P}$ is the physical (real-world) probability measure. The risk-neutral measure will be denoted $\mathbb{Q}$. We recall key definitions from the course as we need them so that this document is self-contained.

Assumption convention. At the start of each section we state explicitly which measure we are working under, which filtration we are conditioning on, and what technical assumptions are in force. This is crucial: the same equation can mean different things under $\mathbb{P}$ and $\mathbb{Q}$.

1 Question 2: Spot rate, forward rate, and the zero-coupon yield curve

1.1 Starting from scratch: why do we need a yield curve at all?

Imagine you are a pension fund manager in January 2026 and you need to invest cash for 10 years. A bank offers you 3% per year. Is that a good deal? You cannot answer that question without knowing what *other* rates are available for different maturities right now. The collection of all such rates – one rate for every maturity T – is what we call the **yield curve** (or the term structure of interest rates). It is, literally, the price of time.

Fixed-income mathematics is built on one primitive object: the **zero-coupon bond**. Everything else – yield curves, forward rates, swaps, swaptions – is ultimately expressed in terms of zero-coupon bond prices. So we start there.

1.2 The zero-coupon bond

A **zero-coupon bond with maturity** T is a contract that pays exactly \$1 at time T and nothing before. There are no coupon payments, no dividends, no intermediate cash flows – just one dollar at one future date. We denote its price at time $t \leq T$ by $P(t, T)$.

Worked Example

Concrete example. Suppose it is time $t = 0$ (today) and a one-year zero-coupon bond trades at $P(0, 1) = 0.95$. This means: to receive \$1.00 in exactly one year, you must pay \$0.95 today. The *cost of waiting* one year is $1/0.95 - 1 \approx 5.26\%$ in simple terms. This is the market's current price for time value over one year.
Now suppose a two-year bond trades at $P(0, 2) = 0.88$. You pay \$0.88 today to receive \$1.00 in two years. The annualised cost of waiting is $(1/0.88)^{1/2} - 1 \approx 6.6\%$.
Notice two things:

1. $P(0, T)$ is *decreasing* in T: the longer you wait, the less you pay today, because your money is locked up longer.
2. $P(t, T) \in (0, 1]$ for $T \geq t$ under positive interest rates: the present value of \$1 in the future is always less than \$1 today.
3. $P(t, t) = 1$ always: a bond that matures right now is worth exactly what it pays, which is \$1.

1.3 The zero-coupon yield $y(t, T)$

Given the bond price $P(t, T)$, we define the **continuously compounded zero-coupon yield** $y(t, T)$ as the constant rate that makes the discounting identity exact:

$$P(t, T) = e^{-y(t,T)(T-t)}.$$

Solving for y:

$$y(t, T) = -\frac{1}{T - t} \log P(t, T).$$

Intuition

Think of $y(t,T)$ as the answer to the question: "*If I could earn a single constant interest rate from t to T, what rate would give me the same bond price?*" It is a *summary statistic* of the entire bond price in one number. When people say "the 10-year yield is 4%", they are referring to this: $y(\text{today}, \text{today} + 10) = 4\%$.

Worked Example

Using our numbers above:

$$y(0,1) = -\log(0.95) \approx 0.0513 = 5.13\%.$$

$$y(0,2) = -\frac{1}{2}\log(0.88) \approx 0.0645 = 6.45\%.$$

Because $y(0,2) > y(0,1)$, the yield curve is *upward sloping*: you demand a higher annualised return for tying your money up for two years instead of one. This is the most common shape in normal economic times, reflecting both expected future rate increases and a term premium for bearing duration risk.

The function $T \mapsto y(t,T)$, plotted for all maturities $T > t$, is the **zero-coupon yield curve at time** t. It encodes everything the market "believes" (under the risk-neutral measure) about future interest rates.

1.4 The instantaneous forward rate $f(t,T)$

The yield $y(t,T)$ is an *average* rate over the whole period $[t,T]$. But sometimes we want to know the *marginal* rate: "what is the market implying for the cost of borrowing for an infinitesimally short interval around a specific future date T?" This is the **instantaneous forward rate** $f(t,T)$.

Formally, it is the negative of the maturity-derivative of the log bond price:

$$f(t,T) := -\frac{\partial}{\partial T}\log P(t,T).$$

Intuition

Here is the best way to think about this. Imagine you are standing at time t and you hold a curve of bond prices $P(t,T)$ for all maturities. The forward curve $T \mapsto f(t,T)$ is like the *slope* of that log-price curve, negated. A steeply declining log-price curve means high forward rates; a flatter curve means low forward rates.
More concretely: $f(t,T)\,\Delta T$ is approximately the rate you would pay, locked in today, for borrowing from time T to time $T + \Delta T$. It is a "futures rate" in continuous time.

Worked Example

Continuing our example, suppose the yield curve is smooth and given by $y(0,T) = 0.04 + 0.01\sqrt{T}$ (just a made-up shape to get numbers). Then

$$\log P(0,T) = -y(0,T)\cdot T = -(0.04T + 0.01T^{3/2}),$$

$$f(0,T) = -\frac{\partial}{\partial T}\log P(0,T) = 0.04 + 0.015\sqrt{T}.$$

At $T = 1$: $f(0,1) = 0.055$. At $T = 4$: $f(0,4) = 0.07$. So the forward curve says: the cost of borrowing for one instant at year 4 is 7%. Compare $y(0,4) = 0.04 + 0.01 \times 2 = 0.06 = 6\%$. The forward rate is *above* the yield at every maturity when the yield curve is upward-sloping. We will see why momentarily.

1.5 Recovering the bond price from the forward curve

Integrating $f(t,T) = -\partial_T \log P(t,T)$ over $[t,T]$ and using $P(t,t) = 1$:

$$\int_t^T f(t,u)\ \mathrm{d}u = -\log P(t,T) + \log P(t,t) = -\log P(t,T).$$

Rearranging:

$$\boxed{P(t,T) = \exp\left(-\int_t^T f(t,u)\ \mathrm{d}u\right).}$$

In plain English: the zero-coupon bond price at time t for maturity T is the exponential of minus the area under the forward curve from t to T. If you integrate all the future instantaneous rates (forward rates) that the market prices today, you get the discount factor. This is exactly analogous to how, in the deterministic case, the discount factor is e^{-rT} for a flat rate r – here the "rate" is not flat but depends on maturity.

1.6 Yield as the average of the forward curve

Substituting the integral representation into the definition of $y(t,T)$:

$$y(t,T) = -\frac{\log P(t,T)}{T-t} = \frac{1}{T-t}\int_t^T f(t,u)\ \mathrm{d}u.$$

In plain English: the zero-coupon yield for maturity T is the simple (maturity-weighted) average of the instantaneous forward rates between t and T. This is why the yield curve is "smoother" than the forward curve: it averages out the peaks and troughs.

Intuition

This averaging relationship is the key to understanding yield curve shapes. When the forward curve is above the yield curve (as in our example), the yield curve is rising. When the forward curve is below the yield curve, the yield curve is falling. When the forward curve crosses the yield curve from above, the yield curve has a local maximum. You can always read the forward curve as "the marginal slope information" hidden inside the yield curve.

1.7 The instantaneous short rate r_t

The **instantaneous short rate** r_t is the limit of the forward rate as the horizon collapses to zero:

$$r_t = \lim_{T\downarrow t} y(t,T) = \lim_{T\downarrow t} f(t,T) = f(t,t).$$

In other words, r_t is the "overnight" (or more precisely, infinitesimally short) rate of return on riskless lending at time t. It is the rate at which the **money-market account** B_t accumulates:

$$B_t = \exp\left(\int_0^t r_s \, \mathrm{d}s\right), \qquad \frac{\mathrm{d}B_t}{B_t} = r_t \, \mathrm{d}t.$$

Unlike $y(t,T)$ and $f(t,T)$, which are observable from market prices, r_t is typically *not* directly observable – it must be modelled or inferred. This is precisely why we need a *short-rate model* (such as CIR, which we study next).

Worked Example

Summary of relationships. Suppose at $t = 0$ the following is observed: $P(0,1) = 0.95$, $P(0,2) = 0.88$, $P(0,3) = 0.80$. Then:

$$\begin{aligned} y(0,1) &= -\log(0.95) \approx 5.13\% \\ y(0,2) &= -\tfrac{1}{2}\log(0.88) \approx 6.45\% \\ y(0,3) &= -\tfrac{1}{3}\log(0.80) \approx 7.44\% \end{aligned}$$

The average forward rate from 0 to 2 equals $y(0,2) = 6.45\%$. The forward rate for the second year alone is obtained by "stripping":

$$P(0,2) = P(0,1) \cdot e^{-f(0;1,2)\cdot 1} \implies f(0;1,2) = \log P(0,1) - \log P(0,2) = \log \frac{0.95}{0.88} \approx 7.7\%.$$

So the market is implying a 7.7% rate for year two. Notice this is higher than both $y(0,1)$ and $y(0,2)$ precisely because the yield curve is upward sloping.

1.8 Static conceptual graph

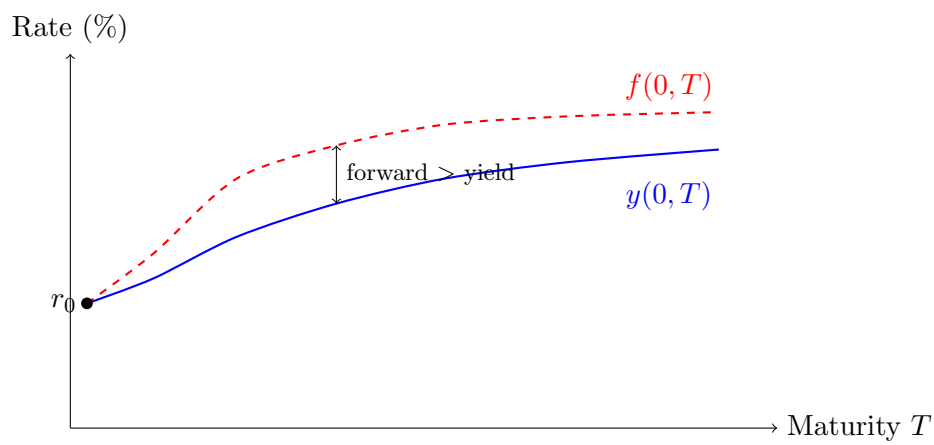

Figure 1: An upward-sloping yield curve and its associated forward curve at $t = 0$. The forward curve lies above the yield curve wherever the yield curve is increasing (because you are pulling the average up from the right). Both curves pass through r_0 at the short end, since for vanishing maturities $y(0,T) \to r_0$ and $f(0,T) \to r_0$. When the yield curve is downward sloping the positions reverse: the forward curve dips below the yield curve.

1.9 Why does the yield curve have a "shape" at all?

A natural question: if we observe yield curves every day, why do they slope up on some days and down on others? There are two forces at work:

1. **Expectations:** If the market believes short rates will rise, long yields must be higher today to compensate (otherwise everyone would park money short-term and roll it over at higher rates).

2. **Risk premium:** Longer bonds are riskier (more sensitive to rate changes), so investors demand a premium – a term premium – for holding them.

Under the physical measure $\mathbb{P}$, both effects contribute. Under the risk-neutral measure $\mathbb{Q}$, the risk premium is absorbed into the drift of r_t, and the yield curve reflects only the $\mathbb{Q}$-expected path of future short rates. We will make this precise in Questions 3 and 4.

1.10 Simulation ideas for iteration 2

- Interactive dashboard: given parameters of a CIR model, plot $y(0,T)$, $f(0,T)$, and $P(0,T)$ as maturity T varies.
- Slider for r_0, $\kappa_{\mathbb{Q}}$, $\theta_{\mathbb{Q}}$, σ showing how the term structure responds.
- Historical: load FRED data for two dates (one normal, one inverted curve) and plot both yield and forward curves side by side.

Open-ended questions

- An *inverted* yield curve (short rates above long rates) has historically preceded recessions. Can you explain why, using the expectations component above?
- What happens to the forward rate $f(t,T)$ as $T \to \infty$? Does it always converge? What does convergence (or divergence) imply about the model?

2 Question 3: Short-rate dynamics under $\mathbb{P}$ (CIR model)

2.1 Recall: the probability space and measure $\mathbb{P}$

Recall from Class

From our course (Definition 5, Reminder sheet), a **probability measure** $\mathbb{P}$ on a measurable space $(\Omega, \mathcal{F})$ is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying $\mathbb{P}(\Omega) = 1$, nonnegativity, and countable additivity. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a **probability space**.
In financial modelling, $\mathbb{P}$ is the **physical** (or **real-world**) measure: it is the probability measure that governs what actually happens in the economy. If you estimated interest rate dynamics from 20 years of daily Fed Funds data and fitted a model, you would be working under $\mathbb{P}$.
From Definition 6, a **filtration** $(\mathcal{F}_t)_{t\geq 0}$ is an increasing family of σ-algebras, with $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. We interpret $\mathcal{F}_t$ as "all information available up to and including time t".

2.2 What is a standard Brownian motion under $\mathbb{P}$?

A **standard Brownian motion under** $\mathbb{P}$, written $W_t^{\mathbb{P}}$, is a continuous-time stochastic process (a family of random variables indexed by time) satisfying:

1. $W_0^{\mathbb{P}} = 0$ (starts at zero),
2. **Independent increments:** for $0 \leq s < t$, $W_t^{\mathbb{P}} - W_s^{\mathbb{P}}$ is independent of $\mathcal{F}_s$ (the future is unpredictable given today),
3. **Gaussian increments:** $W_t^{\mathbb{P}} - W_s^{\mathbb{P}} \sim \mathcal{N}(0, t - s)$ (the increment is normally distributed with variance equal to the elapsed time),
4. **Continuous paths:** the sample paths $t \mapsto W_t^{\mathbb{P}}(\omega)$ are continuous with probability one.

Intuition

Think of $W_t^{\mathbb{P}}$ as the "cumulative surprise" accumulated by time t. At each instant, a tiny shock $\mathrm{d}W_t^{\mathbb{P}} \sim \mathcal{N}(0, \mathrm{d}t)$ arrives. These shocks are independent and have no memory. The variance of the total accumulated shock grows linearly with time: $\mathrm{Var}^{\mathbb{P}}[W_t^{\mathbb{P}}] = t$.
The phrase "under $\mathbb{P}$" tells you which probability measure is generating the distribution. This will matter enormously when we switch to $\mathbb{Q}$: the same process $W_t^{\mathbb{P}}$ will *not* be a Brownian motion under $\mathbb{Q}$ (it will have a drift). This is the content of Girsanov's theorem, which we revisit in Question 4.

2.3 Why do we need a stochastic model for the short rate?

Before we write down the CIR model, let us ask: *why not just use a deterministic interest rate?* After all, central banks announce rates in advance.

The answer is that interest rates are genuinely random over horizons longer than a few weeks. No one in January 2021 predicted that the US Federal Funds rate would go from 0.25% to 5.33% in eighteen months. A deterministic model would price a 10-year bond incorrectly, and it would assign zero value to interest rate derivatives (options on rates), which is clearly wrong. We need a model that:

- Generates a distribution of possible future rates (not just one path),
- Keeps rates non-negative (nominal rates have historically been non-negative in most markets),
- Captures *mean reversion* – rates tend to drift back toward a long-run level rather than wandering off to infinity or zero,
- Is tractable enough to price bonds and derivatives in closed or semi-closed form.

2.4 The CIR square-root diffusion

The **Cox–Ingersoll–Ross (CIR) model** (1985) specifies that the short rate r_t under $\mathbb{P}$ satisfies the stochastic differential equation:

$$\mathrm{d}r_t = \kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\ \mathrm{d}t + \sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{P}}, \qquad r_0 > 0.$$

Let us immediately unpack every piece of this equation, term by term.

The drift term: $\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\ \mathrm{d}t$. This is a **mean-reverting** (Ornstein-Uhlenbeck-type) drift. Let us see why.

- If $r_t < \theta_{\mathbb{P}}$, then $\theta_{\mathbb{P}} - r_t > 0$, so the drift is *positive*: the rate is pulled *upward* toward $\theta_{\mathbb{P}}$.
- If $r_t > \theta_{\mathbb{P}}$, then $\theta_{\mathbb{P}} - r_t < 0$, so the drift is *negative*: the rate is pushed *downward* toward $\theta_{\mathbb{P}}$.
- If $r_t = \theta_{\mathbb{P}}$, the drift is exactly zero: no systematic pull.

The parameter $\kappa_{\mathbb{P}} > 0$ controls *how fast* this pull happens. A large $\kappa_{\mathbb{P}}$ means quick reversion (the rate snaps back rapidly); a small $\kappa_{\mathbb{P}}$ means slow reversion (the rate wanders widely before returning).

The long-run level: $\theta_{\mathbb{P}}$. This is the value toward which r_t is pulled. Economically, it represents the steady-state level of the short rate in the physical world – think of it as the "normal" interest rate environment.

The diffusion term: $\sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{P}}$. This injects randomness. The key feature is that the volatility is **state-dependent**: it is proportional to $\sqrt{r_t}$.

Why $\sqrt{r_t}$ and not just a constant σ? Two reasons:

1. When r_t is close to zero, $\sqrt{r_t} \approx 0$, so the stochastic part is very small. Combined with the positive drift $\kappa_{\mathbb{P}}\theta_{\mathbb{P}}$ at the boundary, this keeps $r_t \geq 0$.
2. Empirically, interest rate volatility *does* tend to be higher when rates are higher (e.g., the volatile high-rate environments of the 1980s). The $\sqrt{r_t}$ specification captures this heteroscedasticity.

Worked Example

Intuitive parameter choices. Suppose we calibrate to US data and find $\kappa_{\mathbb{P}} = 0.3$, $\theta_{\mathbb{P}} = 0.04$ (4%), $\sigma = 0.05$, $r_0 = 0.02$ (2%).

- **Half-life of reversion:** The mean reversion half-life is $\log(2)/\kappa_{\mathbb{P}} = \log(2)/0.3 \approx 2.3$

years. So if rates start at 2%, they will be approximately halfway back to 4% after 2.3 years on average.

- **Instantaneous volatility at $r_0 = 0.02$:** $\sigma\sqrt{r_0} = 0.05\sqrt{0.02} \approx 0.007$ per year, i.e., 70 basis points per year of standard deviation.
- **Volatility at $r = 0.08$:** $0.05\sqrt{0.08} \approx 0.014$, double the volatility at 2%. Higher rates come with more noise, which is realistic.

2.5 (i) Why does this model admit a strong solution?

Weak vs. strong solutions. There are two notions of "solution" to an SDE $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t$.

- A **weak solution** is a triplet (probability space, Brownian motion, adapted process) satisfying the SDE. You get to *choose* the probability space and Brownian motion.
- A **strong solution** is an adapted process built on a *given* probability space with a *given* Brownian motion, satisfying the SDE pathwise (i.e., almost surely for every ω). You must *work with the Brownian motion you are handed.*

A strong solution is more demanding. If a strong solution exists and is unique (pathwise uniqueness), we say the SDE is *well-posed.*

Standard sufficient conditions. The classical result (Itô, 1951) is: if b and σ are globally **Lipschitz continuous** and have **linear growth**, then there exists a unique strong solution. For CIR:

- Drift: $b(r) = \kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r)$. This is linear in r, hence globally Lipschitz. Check: $|b(r_1) - b(r_2)| = \kappa_{\mathbb{P}}|r_1 - r_2|$.
- Diffusion: $a(r) = \sigma\sqrt{r}$. This is **Hölder-$\frac{1}{2}$** but *not* Lipschitz near $r = 0$, because $|\sqrt{r_1} - \sqrt{r_2}| \leq \sqrt{|r_1 - r_2|}$, which is sublinear but not linear.

So the standard Lipschitz theorem does not directly apply. However, there is a more refined result by **Yamada and Watanabe (1971)** that extends pathwise uniqueness to diffusion coefficients of the form $\sigma(r) = |r|^\alpha$ for $\alpha \geq 1/2$. Since our diffusion is exactly $\sigma\sqrt{r}$ (i.e., $\alpha = 1/2$), the Yamada–Watanabe theorem guarantees pathwise uniqueness. Combined with the fact that weak existence holds (the drift and diffusion have linear growth), we obtain existence and uniqueness of a **unique strong solution** by the Yamada-Watanabe principle.

Furthermore, this strong solution satisfies $r_t \geq 0$ for all $t \geq 0$ almost surely, because the Feller boundary condition (see next section) prevents the process from going negative.

2.6 (ii) Parameter constraints: the Feller condition

Three basic conditions for the model to make sense:

$$\kappa_{\mathbb{P}} > 0, \qquad \theta_{\mathbb{P}} > 0, \qquad \sigma > 0, \qquad r_0 \geq 0.$$

These ensure mean reversion toward a positive level with positive randomness.

Beyond these, the key constraint is the **Feller condition**:

$$2\kappa_{\mathbb{P}}\theta_{\mathbb{P}} \geq \sigma^2.$$

Where does this come from? The boundary $r = 0$ is special for the CIR process. When r_t is near zero, the diffusion coefficient $\sigma\sqrt{r_t} \approx 0$ vanishes, but the drift $\kappa_{\mathbb{P}}\theta_{\mathbb{P}} > 0$ provides a positive push away from zero. The Feller condition says: *the upward drift at zero is strong enough relative to the downward noise pressure that the process cannot reach zero.*

More precisely:

- If $2\kappa_{\mathbb{P}}\theta_{\mathbb{P}} > \sigma^2$ (**strict Feller**): the boundary $r = 0$ is **inaccessible** – the process stays strictly positive with probability 1.
- If $2\kappa_{\mathbb{P}}\theta_{\mathbb{P}} = \sigma^2$: the boundary is accessible but **instantaneously reflecting** – r_t can touch zero but immediately bounces back.
- If $2\kappa_{\mathbb{P}}\theta_{\mathbb{P}} < \sigma^2$ (**violation**): the boundary is accessible and the process can *spend positive time* at zero (but it remains non-negative; it does not go below zero).

Important: violating the Feller condition does *not* destroy the strong solution. The process r_t is still well-defined, non-negative, and the CIR formula remains valid. It merely changes the behavior at the zero boundary.

2.7 (iii) Interpreting the constraints with examples

Let us fix $\sigma = 0.10$ and vary κ and θ to see the Feller condition at work.

$\kappa_{\mathbb{P}}$	$\theta_{\mathbb{P}}$	$2\kappa_{\mathbb{P}}\theta_{\mathbb{P}}$	σ^2	Status
0.5	0.10	0.100	0.010	Strict Feller satisfied (10× margin)
0.5	0.01	0.010	0.010	Exactly on boundary
0.5	0.005	0.005	0.010	Feller *violated*
0.1	0.05	0.010	0.010	Exactly on boundary

- In row 3, $\theta_{\mathbb{P}} = 0.005$ (the long-run rate is only 50 basis points). The noise level ($\sigma = 0.10$) is too large for such a low target – the process will occasionally touch zero. This is not catastrophic for the model's mathematical well-posedness, but it is economically suspect: a rate of exactly zero corresponds to an extreme liquidity trap, and the model should not spend substantial time there.
- In row 1, there is a 10-fold margin of safety. This corresponds to a "normal" regime where the long-run rate (10%) is high enough and reversion strong enough that near-zero rates are very rare.

2.8 (iv) The distribution of r_t

One of the most remarkable features of the CIR model is that we can compute the *exact* distribution of r_t for any $t > 0$.

The result. Conditional on r_0, the distribution of r_t is a **scaled non-central chi-squared distribution**:

$$r_t \stackrel{d}{=} c_t \cdot X, \qquad X \sim \chi^2_\nu(\lambda_t),$$

where

$$c_t = \frac{\sigma^2(1-e^{-\kappa_{\mathbb{P}} t})}{4\kappa_{\mathbb{P}}}, \qquad \nu = \frac{4\kappa_{\mathbb{P}}\theta_{\mathbb{P}}}{\sigma^2}, \qquad \lambda_t = \frac{4\kappa_{\mathbb{P}} e^{-\kappa_{\mathbb{P}} t}\, r_0}{\sigma^2(1-e^{-\kappa_{\mathbb{P}} t})}.$$

Here ν is the **degrees of freedom** and λ_t is the **non-centrality parameter**.

What is a non-central chi-squared distribution? You already know that if $Z_1, \ldots, Z_k$ are independent $\mathcal{N}(0,1)$ variables, then $Z_1^2 + \cdots + Z_k^2 \sim \chi^2_k$ (central chi-squared). The *non-central* version $\chi^2_k(\lambda)$ arises when the Z_i have non-zero means: if $Z_i \sim \mathcal{N}(\mu_i, 1)$ then $\sum Z_i^2 \sim \chi^2_k(\lambda)$ with $\lambda = \sum \mu_i^2$. The non-centrality parameter shifts the distribution to the right.

For CIR, the intuition is:

- ν measures the "number of degrees of freedom", which encodes how much the long-run level $\theta_{\mathbb{P}}$ and mean-reversion $\kappa_{\mathbb{P}}$ pull the distribution away from zero. Under strict Feller ($\nu > 2$), the distribution has a proper interior mode (does not pile up at zero).
- λ_t measures how far the current rate r_0 contributes to the distribution. As $t \to \infty$, $\lambda_t \to 0$ (the memory of the starting point fades).
- c_t is a scaling factor that grows from 0 at $t = 0$ to $\sigma^2/(4\kappa_{\mathbb{P}})$ as $t \to \infty$.

Moments. From the known moments of the non-central chi-squared:

$$\mathbb{E}^{\mathbb{P}}[r_t] = \theta_{\mathbb{P}} + (r_0 - \theta_{\mathbb{P}})e^{-\kappa_{\mathbb{P}} t}.$$

In English: the expected short rate at time t is a weighted average of its current level r_0 and its long-run level $\theta_{\mathbb{P}}$, with the weight on r_0 decaying exponentially at speed $\kappa_{\mathbb{P}}$. As $t \to \infty$, $\mathbb{E}^{\mathbb{P}}[r_t] \to \theta_{\mathbb{P}}$. This is the mean reversion in expectation.

$$\mathrm{Var}^{\mathbb{P}}(r_t) = \frac{\sigma^2 r_0}{\kappa_{\mathbb{P}}} e^{-\kappa_{\mathbb{P}} t}(1-e^{-\kappa_{\mathbb{P}} t}) + \frac{\sigma^2\theta_{\mathbb{P}}}{2\kappa_{\mathbb{P}}}(1-e^{-\kappa_{\mathbb{P}} t})^2.$$

At short horizons, variance is approximately $\sigma^2 r_0 \cdot t$ (linear in time, as for Brownian motion). At long horizons, variance saturates at $\sigma^2\theta_{\mathbb{P}}/(2\kappa_{\mathbb{P}})$ – the stationary variance.

Worked Example

With $\kappa_{\mathbb{P}} = 0.3$, $\theta_{\mathbb{P}} = 0.04$, $\sigma = 0.05$, $r_0 = 0.02$:

$$\mathbb{E}^{\mathbb{P}}[r_1] = 0.04 + (0.02 - 0.04)e^{-0.3} \approx 0.04 - 0.0148 = 2.52\%.$$

After one year, the rate has drifted about halfway toward the 4% mean. The variance:

$$\mathrm{Var}^{\mathbb{P}}(r_1) \approx \frac{0.0025 \times 0.02}{0.3}(1-e^{-0.3}) \cdot e^{-0.3} + \frac{0.0025 \times 0.04}{0.6}(1-e^{-0.3})^2 \approx 4.8 \times 10^{-5}.$$

Standard deviation $\approx 0.69\%$, so a 95% interval for r_1 is roughly $2.52\% \pm 1.35\%$, i.e., between 1.17% and 3.87%.

2.9 (v) Mathematical properties and why CIR is appropriate

Let us state each key property and explain why it matters for modelling interest rates.

1. Nonnegativity. $r_t \geq 0$ for all t. Interest rates are non-negative in standard nominal terms (ignoring the ECB/BoJ near-negative rate era, which requires model extensions). A model that allows $r_t < 0$ would imply a money-market account that *shrinks* over time, which is unrealistic in standard theory.

2. Mean reversion. Rates revert to $\theta_{\mathbb{P}}$ at speed $\kappa_{\mathbb{P}}$. This is economically essential: central banks actively manage rates toward policy targets. A random walk ($\kappa = 0$) for interest rates would imply unbounded long-run variance – the rate could drift to 100% or to negative infinity – which is empirically implausible. Mean reversion keeps the distribution stationary.

3. State-dependent (heteroscedastic) volatility. $\mathrm{Vol}(r_t) = \sigma\sqrt{r_t}$. Empirically, periods of high interest rates have shown higher volatility. This is captured automatically by the CIR specification.

4. Markov property. Given r_t, the future evolution of r does not depend on the history $(r_s)_{s<t}$. Mathematically: r_t is a Markov process. This property is what allows us to price bonds by solving a PDE in just two variables (t, r), rather than an infinite-dimensional path-dependent problem.

5. Affine structure. This is the most important property for pricing. A short-rate model is called **affine** if the conditional Laplace transform of r_t (or more generally of $\int_t^T r_s \, \mathrm{d}s$) is an exponential-affine function of r_t:

$$\mathbb{E}_t^{\mathbb{P}}\left[\exp\left(-u \int_t^T r_s \, \mathrm{d}s\right)\right] = \exp\big(A(t,T;u) - B(t,T;u)\, r_t\big),$$

for deterministic functions A and B. This is what gives us the log-affine bond pricing formula we derive in Question 6.

6. Stationary distribution. As $t \to \infty$, r_t converges in distribution to a **Gamma distribution**:

$$r_\infty \sim \mathrm{Gamma}\left(\frac{2\kappa_{\mathbb{P}}\theta_{\mathbb{P}}}{\sigma^2}, \frac{\sigma^2}{2\kappa_{\mathbb{P}}}\right).$$

This gives a natural prior for long-run rate levels: skewed right, bounded below by zero.

Worked Example

With $\kappa_{\mathbb{P}} = 0.3$, $\theta_{\mathbb{P}} = 0.04$, $\sigma = 0.05$:

$$\text{Shape} = \frac{2 \times 0.3 \times 0.04}{0.05^2} = 9.6, \quad \text{Scale} = \frac{0.05^2}{2 \times 0.3} \approx 0.00417.$$

So the stationary mean is $9.6 \times 0.00417 \approx 4\% = \theta_{\mathbb{P}}$ (as expected), and the stationary standard deviation is $\sqrt{9.6} \times 0.00417 \approx 1.29\%$. Rates will hover around 4% with a standard deviation of about 130 basis points in the long run.

2.10 Static conceptual graph

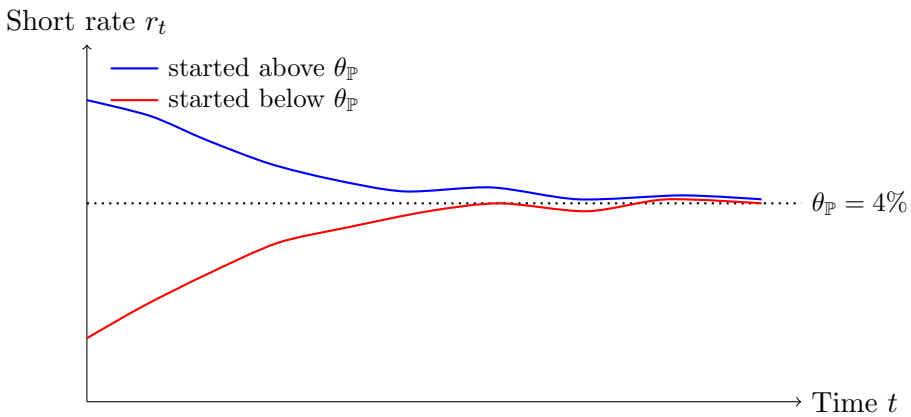

Figure 2: Qualitative CIR sample paths. Both paths revert toward $\theta_{\mathbb{P}}$ (dashed line), but with stochastic fluctuations. Paths starting above $\theta_{\mathbb{P}}$ drift downward; those starting below drift upward. Volatility is lower near the origin because of the $\sqrt{r_t}$ factor.

Open-ended questions

- The CIR model has only one factor (r_t). Real yield curves move in three dimensions (level, slope, curvature). How would a two-factor extension look, and what would the second factor represent?
- In recent years (2014-2021 in Europe), short rates were genuinely negative. What does that imply about the CIR model as an empirical description of those markets?

3 Question 4: Risk-neutral dynamics under $\mathbb{Q}$

3.1 Why do we need a different measure?

We have described r_t under $\mathbb{P}$ – the measure that reflects actual probabilities. But pricing a bond requires computing an expectation of a future payoff. The question is: under *which* probability measure?

The answer from no-arbitrage theory – which we established in class through the first fundamental theorem of asset pricing – is that prices of traded assets must be *martingales under some equivalent probability measure* $\mathbb{Q}$ (after discounting by the money-market account).

Recall from Class

From Definition 14 (Reminder sheet), a process M_t is a $\mathbb{Q}$-**martingale** if $\mathbb{E}_s^{\mathbb{Q}}[M_t] = M_s$ for $s < t$.
From Definition 23, a **martingale measure** (risk-neutral measure) $\mathbb{Q}$ is an equivalent probability measure under which every discounted traded asset is a martingale.
"Equivalent" means $\mathbb{Q}$ and $\mathbb{P}$ agree on which events are possible (zero-probability sets coincide), but they assign different probabilities to non-null events.

3.2 The Radon-Nikodym derivative: the "exchange rate" between measures

Since $\mathbb{Q}$ and $\mathbb{P}$ are equivalent, there exists a positive random variable $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$ (the **Radon-Nikodym derivative**) such that for any random variable X:

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}\left[X \cdot \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right].$$

Think of $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(\omega)$ as the "weight" that adjusts $\mathbb{P}$-probabilities to $\mathbb{Q}$-probabilities at each scenario ω. Scenarios where $\mathbb{Q}$ assigns more probability than $\mathbb{P}$ have weight > 1; scenarios where $\mathbb{Q}$ assigns less have weight < 1.

3.3 Girsanov's theorem: the bridge between $\mathbb{P}$ and $\mathbb{Q}$

Girsanov's theorem (in continuous time) says: given a process $(\lambda_t)_{t\geq 0}$ satisfying certain integrability conditions, we can define a new measure $\mathbb{Q}$ via:

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T, \quad Z_t = \exp\left(-\int_0^t \lambda_s \, \mathrm{d}W_s^{\mathbb{P}} - \tfrac{1}{2}\int_0^t \lambda_s^2 \, \mathrm{d}s\right).$$

Under $\mathbb{Q}$, the process

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + \int_0^t \lambda_s \, \mathrm{d}s$$

is a standard Brownian motion.

In English: we take the $\mathbb{P}$-Brownian motion and add a drift to it. The new process $W_t^{\mathbb{Q}}$ is a genuine Brownian motion under $\mathbb{Q}$, but from the $\mathbb{P}$ perspective it has a drift $-\lambda_t \, \mathrm{d}t$ subtracted. The parameter λ_t is called the **market price of risk**: it tells you how much drift the market demands for bearing one unit of Brownian risk.

3.4 From $\mathbb{P}$ to $\mathbb{Q}$: rewriting the CIR SDE

Under $\mathbb{P}$:

$$\mathrm{d}r_t = \kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\ \mathrm{d}t + \sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{P}}.$$

Since $\mathrm{d}W_t^{\mathbb{P}} = \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t\ \mathrm{d}t$ (from Girsanov):

$$\mathrm{d}r_t = \kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\ \mathrm{d}t + \sigma\sqrt{r_t}(\ \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t\ \mathrm{d}t) = \left[\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t) - \sigma\sqrt{r_t}\lambda_t\right]\mathrm{d}t + \sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{Q}}.$$

We want this to look like a CIR model under $\mathbb{Q}$:

$$\mathrm{d}r_t = \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\ \mathrm{d}t + \sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{Q}}.$$

(Note: the σ is the same under both measures – Girsanov only changes the drift, not the diffusion coefficient.)

3.5 (ii) The required functional form of λ_t

For the $\mathbb{Q}$-drift to be of the form $\kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)$ (affine in r_t), we need

$$\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t) - \sigma\sqrt{r_t}\lambda_t = \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} - \kappa_{\mathbb{Q}}r_t.$$

Rearranging:

$$\sigma\sqrt{r_t}\lambda_t = (\kappa_{\mathbb{P}} - \kappa_{\mathbb{Q}})r_t + (\kappa_{\mathbb{P}}\theta_{\mathbb{P}} - \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}}).$$

The right-hand side must be expressible as $\sigma\sqrt{r_t}\lambda_t$. For this to hold for all $r_t > 0$, we need λ_t of the form:

$$\lambda_t = \frac{(\kappa_{\mathbb{P}} - \kappa_{\mathbb{Q}})r_t + (\kappa_{\mathbb{P}}\theta_{\mathbb{P}} - \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}})}{\sigma\sqrt{r_t}} = \underbrace{\frac{\kappa_{\mathbb{P}}\theta_{\mathbb{P}} - \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}}}{\sigma\sqrt{r_t}}}_{\text{term 1}} + \underbrace{\frac{\kappa_{\mathbb{P}} - \kappa_{\mathbb{Q}}}{\sigma}\sqrt{r_t}}_{\text{term 2}}.$$

The most common special case (and the standard CIR market price of risk) sets the term 1 to zero (to avoid singularity at $r_t = 0$):

$$\kappa_{\mathbb{P}}\theta_{\mathbb{P}} = \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}}, \quad \lambda_t = \frac{\kappa_{\mathbb{P}} - \kappa_{\mathbb{Q}}}{\sigma}\sqrt{r_t} =: \eta\sqrt{r_t}.$$

In this case:

$$\kappa_{\mathbb{Q}} = \kappa_{\mathbb{P}} + \sigma\eta, \qquad \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} = \kappa_{\mathbb{P}}\theta_{\mathbb{P}}.$$

Intuition

The market price of risk $\lambda_t = \eta\sqrt{r_t}$ says: *the compensation for bearing interest rate risk is proportional to the square root of the current rate.* When rates are high (stressful times for bondholders), the market demands a higher risk premium. When rates are near zero, the risk premium is also near zero (there is little systematic risk in a near-zero rate environment). This state-dependence is what makes the CIR risk premium economically sensible.

3.6 (i) The link between $(\kappa_{\mathbb{P}}, \theta_{\mathbb{P}})$ and $(\kappa_{\mathbb{Q}}, \theta_{\mathbb{Q}})$

Collecting the results:

$$\boxed{\kappa_{\mathbb{Q}} = \kappa_{\mathbb{P}} + \sigma\eta, \qquad \theta_{\mathbb{Q}} = \frac{\kappa_{\mathbb{P}}\theta_{\mathbb{P}}}{\kappa_{\mathbb{P}} + \sigma\eta}.}$$

Several important observations:

1. If $\eta > 0$ (risk aversion: investors accept lower returns for safety), then $\kappa_{\mathbb{Q}} > \kappa_{\mathbb{P}}$: mean reversion is *faster* under $\mathbb{Q}$ than under $\mathbb{P}$. Under $\mathbb{Q}$, rates appear to pull toward the long-run level more quickly because the risk-neutral pricing already "looks through" the risk premium.

2. The long-run levels satisfy: $\kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} = \kappa_{\mathbb{P}}\theta_{\mathbb{P}}$ (constant product). So if $\kappa_{\mathbb{Q}} > \kappa_{\mathbb{P}}$ then $\theta_{\mathbb{Q}} < \theta_{\mathbb{P}}$: the risk-neutral long-run level is *lower* than the physical one. This is the risk premium compressed into the long end of the yield curve.

3. η is called the **price of risk** coefficient. It is *not* directly observable from market data. To identify it, you need *both* historical time series data (to get $\kappa_{\mathbb{P}}, \theta_{\mathbb{P}}$) *and* bond or derivative prices (to get $\kappa_{\mathbb{Q}}, \theta_{\mathbb{Q}}$).

Worked Example

If $\kappa_{\mathbb{P}} = 0.3$, $\theta_{\mathbb{P}} = 0.04$, $\sigma = 0.05$, and $\eta = 1.0$:

$$\kappa_{\mathbb{Q}} = 0.3 + 0.05 \times 1.0 = 0.35, \quad \theta_{\mathbb{Q}} = \frac{0.3 \times 0.04}{0.35} \approx 0.0343 = 3.43\%.$$

So the risk-neutral long-run level is 3.43%, which is below the physical 4%. The difference (57 basis points) represents the term premium embedded in long-maturity yields. This is why the risk-neutral yield curve tends to be "flatter" than the physical expectations component: the risk premium adds slope under $\mathbb{P}$ that disappears under $\mathbb{Q}$.

3.7 (iii) Proving the Novikov condition

What is the Novikov condition and why does it matter? We defined the Radon-Nikodym derivative Z_T earlier. For Z_T to define a *valid* probability measure $\mathbb{Q}$, we need Z_T to be a true martingale (not just a local martingale). If Z_T is only a local martingale, then $\mathbb{E}^{\mathbb{P}}[Z_T]$ could be strictly less than 1, meaning "probability mass is lost" and $\mathbb{Q}$ is not a proper probability measure.

The **Novikov condition** (Novikov, 1972) provides a simple sufficient condition:

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \lambda_t^2 \, \mathrm{d}t\right)\right] < \infty.$$

If this holds, then Z_t is a true martingale and $\mathbb{Q}$ is a valid probability measure.

Why is Novikov checking the "right" thing? The process Z_t involves a stochastic integral $\int_0^t \lambda_s \, \mathrm{d}W_s^{\mathbb{P}}$ plus a compensating drift $-\frac{1}{2}\int_0^t \lambda_s^2 \, \mathrm{d}s$. The stochastic integral part is automatically a local martingale. The issue is whether it is a true (global) martingale. Novikov's condition checks that the "size" of the integrand λ_t (measured by $\int_0^T \lambda_t^2 \, \mathrm{d}t$) is not so large that probability mass leaks to infinity.

Verifying Novikov for $\lambda_t = \eta\sqrt{r_t}$. With this choice:

$$\frac{1}{2}\int_0^T \lambda_t^2 \, \mathrm{d}t = \frac{\eta^2}{2}\int_0^T r_t \, \mathrm{d}t.$$

We need to show:

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{\eta^2}{2}\int_0^T r_t \, \mathrm{d}t\right)\right] < \infty.$$

The CIR process is an **affine process**, which means that its moment generating function for functionals of the form $\int_0^T r_t \, \mathrm{d}t$ is exponential-affine in r_0:

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(u\int_0^T r_t \, \mathrm{d}t\right)\right] = \exp\big(\alpha(T;u) + \beta(T;u)\, r_0\big),$$

provided the Riccati ODEs for (α, β) have a solution on $[0, T]$. This solution exists for $u < u^\star$ where $u^\star$ depends on $\kappa_{\mathbb{P}}$ and σ. Since $u = \eta^2/2$ is fixed, for any finite T and reasonable η the Novikov integral is finite. Therefore the Novikov condition is satisfied on finite horizons.

Intuition

In plain terms: since r_t is a non-negative, mean-reverting process that cannot grow without bound, the integral $\int_0^T r_t \, \mathrm{d}t$ is a.s. finite and cannot explode. The exponential of this integral is finite in expectation because the tails of the CIR distribution are sub-Gaussian for reasonable parameters. Contrast this with a process that can run away to infinity, for which Novikov could fail.

3.8 (iv) Why maintain the same structural specification under $\mathbb{Q}$?

First: what does "not directly observable under $\mathbb{Q}$" mean? The measure $\mathbb{Q}$ is a mathematical construction: it is not what you observe when you watch the economy. Time series of bond prices and rate data are generated by the real world – by $\mathbb{P}$. The risk-neutral measure $\mathbb{Q}$ exists as a pricing tool, but there is no direct way to estimate $\mathbb{Q}$-probabilities from historical data.

To see this, consider the following. Under $\mathbb{Q}$, the expected return on every asset equals the risk-free rate r_t (that is the definition of $\mathbb{Q}$). But in reality, riskier assets earn higher expected returns under $\mathbb{P}$ (the equity premium, the term premium, etc.). So if you tried to use historical average returns to infer $\mathbb{Q}$-expected returns, you would get the wrong answer.

So why keep the same CIR structure? The compelling reason is **parsimony and tractability**:

1. **One state variable, two measures.** There is one object that drives interest rates: the short rate r_t. It exists in the real world and is observed (approximately, through LIBOR, Fed Funds, OIS, etc.). Under $\mathbb{P}$ and $\mathbb{Q}$ it follows the same type of diffusion (CIR), just with different drift parameters. This is elegant and consistent.

2. **Closed-form bond prices.** The CIR structure under $\mathbb{Q}$ (affine drift, square-root diffusion) is exactly what allows us to derive the log-affine bond pricing formula in Question 6. If we changed the structure under $\mathbb{Q}$ (e.g., made the diffusion coefficient σr_t instead of $\sigma\sqrt{r_t}$), we would lose tractability.

3. **Stable calibration.** Keeping the CIR form under $\mathbb{Q}$ means we have only three parameters $(\kappa_{\mathbb{Q}}, \theta_{\mathbb{Q}}, \sigma)$ to calibrate to cross-sectional bond prices. This is manageable.

4. **Theoretical coherence.** The Girsanov change only changes the drift, not the diffusion. So if you start with a CIR diffusion $(\sigma\sqrt{r_t})$ under $\mathbb{P}$, it is *automatic* that the diffusion remains $\sigma\sqrt{r_t}$ under $\mathbb{Q}$ – only the drift $\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)$ changes. Maintaining the same structural form is not an assumption – it is a consequence of the specific market price of risk $\lambda_t = \eta\sqrt{r_t}$.

Effect of the measure change on expectations and variance. The change from $\mathbb{P}$ to $\mathbb{Q}$ affects the drift of r_t but not its diffusion:

$$\mathbb{E}^{\mathbb{Q}}[r_t \mid r_0] = \theta_{\mathbb{Q}} + (r_0 - \theta_{\mathbb{Q}})e^{-\kappa_{\mathbb{Q}}t} \quad \text{(different from } \mathbb{P} \text{ expectation)},$$
$$\mathrm{Var}^{\mathbb{Q}}(r_t) = \frac{\sigma^2 r_0}{\kappa_{\mathbb{Q}}} e^{-\kappa_{\mathbb{Q}}t}(1 - e^{-\kappa_{\mathbb{Q}}t}) + \frac{\sigma^2\theta_{\mathbb{Q}}}{2\kappa_{\mathbb{Q}}}(1 - e^{-\kappa_{\mathbb{Q}}t})^2 \quad \text{(different scale, same structure)}.$$

The *local* variance (the diffusion coefficient squared times $\mathrm{d}t$) $\sigma^2 r_t \, \mathrm{d}t$ is the same under both measures. Only the *conditional mean* changes.

Open-ended questions

- If $\eta < 0$ (risk-seeking behaviour, or unusual environments like post-2008), what happens to $\kappa_{\mathbb{Q}}$ and $\theta_{\mathbb{Q}}$? Does the Feller condition under $\mathbb{Q}$ survive?
- How would you estimate η empirically? What data would you use?

4 Question 5: The discounted bond price is a $\mathbb{Q}$-martingale

4.1 Setup and what we want to prove

Recall from Class

From Definition 14 (Reminder sheet), a process M is a $\mathbb{Q}$-**martingale** on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{Q})$ if:

1. $\mathbb{E}^{\mathbb{Q}}[|M_t|] < \infty$ for all t,
2. M_t is $\mathcal{F}_t$-measurable (adapted) for all t,
3. For all $s \leq t$: $\mathbb{E}^{\mathbb{Q}}[M_t \mid \mathcal{F}_s] = M_s$.

The third property is the key one: the best forecast of a martingale's future value, given current information, is its current value. There is no predictable drift.

We want to show that the **discounted bond price**

$$\widetilde{P}(t,T) := \frac{P(t,T)}{B_t} = P(t,T)\exp\left(-\int_0^t r_s \, \mathrm{d}s\right)$$

is a $\mathbb{Q}$-martingale.

4.2 Why this should be true: the no-arbitrage argument

Before the formal proof, let us understand why this must hold.

B_t is the money-market account: if you invest \$1 at time 0 and earn the risk-free short rate continuously, you accumulate B_t by time t. It is the "benchmark" or "numeraire".

If $\widetilde{P}(t,T) = P(t,T)/B_t$ were *not* a martingale under $\mathbb{Q}$, there would exist times $s < t$ for which $\mathbb{E}^{\mathbb{Q}}[\widetilde{P}(t,T) \mid \mathcal{F}_s] \neq \widetilde{P}(s,T)$. But this would mean:

- If $\mathbb{E}^{\mathbb{Q}}[\widetilde{P}(t,T) \mid \mathcal{F}_s] > \widetilde{P}(s,T)$: the discounted bond "earns more than the risk-free rate" on average under $\mathbb{Q}$ – an arbitrage.
- If $\mathbb{E}^{\mathbb{Q}}[\widetilde{P}(t,T) \mid \mathcal{F}_s] < \widetilde{P}(s,T)$: the discounted bond "earns less than the risk-free rate" – you would short it and earn a riskless profit.

So the martingale property is equivalent to the no-arbitrage condition for the bond market.

4.3 The proof

Step 1: Write the bond price as a risk-neutral expectation.

By the risk-neutral pricing formula (the definition of $\mathbb{Q}$ in this context):

$$P(t,T) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u \, \mathrm{d}u\right)\right] = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u \, \mathrm{d}u\right) \middle| \mathcal{F}_t\right].$$

This says: the bond price at t is the $\mathbb{Q}$-expected value of the discount factor $e^{-\int_t^T r_u \, \mathrm{d}u}$ (the "stochastic discount factor" from t to T), given all information up to t.

Step 2: Write the discounted bond price.

Multiply both sides by $B_t^{-1} = e^{-\int_0^t r_u \, \mathrm{d}u}$:

$$\widetilde{P}(t,T) = e^{-\int_0^t r_u \, \mathrm{d}u} \cdot \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_u \, \mathrm{d}u} \middle| \mathcal{F}_t \right].$$

Since $e^{-\int_0^t r_u \, \mathrm{d}u}$ is $\mathcal{F}_t$-measurable (it depends only on information up to t), we can pull it inside the conditional expectation (using property EC6 from the class notes: if Y is $\mathcal{F}_t$-measurable, then $\mathbb{E}^{\mathbb{Q}}[XY|\mathcal{F}_t] = Y\mathbb{E}^{\mathbb{Q}}[X|\mathcal{F}_t]$):

$$\widetilde{P}(t,T) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^t r_u \, \mathrm{d}u} \cdot e^{-\int_t^T r_u \, \mathrm{d}u} \middle| \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r_u \, \mathrm{d}u} \middle| \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}}\left[B_T^{-1} \middle| \mathcal{F}_t \right].$$

Step 3: Apply the tower property.
Now take $s \leq t$ and compute $\mathbb{E}^{\mathbb{Q}}[\widetilde{P}(t,T) \mid \mathcal{F}_s]$:

$$\mathbb{E}^{\mathbb{Q}}\left[\widetilde{P}(t,T) \middle| \mathcal{F}_s \right] = \mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}\left[B_T^{-1} \middle| \mathcal{F}_t \right] \middle| \mathcal{F}_s \right].$$

Applying the **tower property of conditional expectation** (property EC3 from the Reminder: $\mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}[X|\mathcal{F}_t]|\mathcal{F}_s] = \mathbb{E}^{\mathbb{P}}[X|\mathcal{F}_s]$ when $s \leq t$, and the same holds under $\mathbb{Q}$):

$$\mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}\left[B_T^{-1} \middle| \mathcal{F}_t \right] \middle| \mathcal{F}_s \right] = \mathbb{E}^{\mathbb{Q}}\left[B_T^{-1} \middle| \mathcal{F}_s \right] = \widetilde{P}(s,T).$$

Conclusion: For all $0 \leq s \leq t \leq T$,

$$\mathbb{E}^{\mathbb{Q}}\left[\widetilde{P}(t,T) \middle| \mathcal{F}_s \right] = \widetilde{P}(s,T).$$

This is exactly the martingale property (condition M3 from Definition 14). The discounted bond price is a $\mathbb{Q}$-martingale. $\square$

4.4 Financial interpretation of each step

- **Step 1** expresses the fundamental risk-neutral pricing principle: prices are $\mathbb{Q}$-expectations of discounted payoffs. Under $\mathbb{Q}$, there is no risk premium – all assets grow at the risk-free rate in expectation.
- **Step 2** combines the discount factor from $[0,t]$ (which we already know) with the stochastic discount factor from $[t,T]$ (which is still random at time s). The product $e^{-\int_0^T r_u \, \mathrm{d}u}$ is the total discount from 0 to T.
- **Step 3** is the key probability-theoretic step: the tower property says "if you first compute the best estimate given $\mathcal{F}_t$ and then refine it given $\mathcal{F}_s \subseteq \mathcal{F}_t$, you get the same as directly computing the best estimate given $\mathcal{F}_s$". This is the iterated expectations principle from your class notes.

4.5 The SDE form of the martingale

Since $\widetilde{P}$ is a $\mathbb{Q}$-martingale and is continuous, it can be written as a stochastic integral (by the Martingale Representation Theorem):

$$\mathrm{d}\widetilde{P}(t,T) = \widetilde{P}(t,T) \cdot \Sigma(t,T) \, \mathrm{d}W_t^{\mathbb{Q}}$$

for some adapted process $\Sigma(t,T)$ (the "bond volatility"). The drift is zero, confirming the martingale property. We will compute $\Sigma(t,T)$ explicitly in Question 6.

Open-ended questions

- Can you give a direct proof (without going through the risk-neutral pricing formula) using the SDE for $P(t, T)$ derived in Question 6?
- What changes in the proof if r_t can be negative?

5 Question 6: Zero-coupon bond price in log-affine form (CIR)

5.1 The pricing formula and what we are computing

We want to find the explicit formula for:

$$P(t,T) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s \, \mathrm{d}s\right)\right].$$

Let us carefully unpack this before computing it.

- $\exp\left(-\int_t^T r_s \, \mathrm{d}s\right)$ is the **stochastic discount factor**: it discounts \$1 paid at T back to time t using the random short rate path. If rates are high over $[t,T]$, this factor is small (heavy discounting). If rates are low, it is closer to 1.
- $\mathbb{E}_t^{\mathbb{Q}}[\cdot]$ means we are taking the expectation under the risk-neutral measure $\mathbb{Q}$, conditional on $\mathcal{F}_t$ (everything known at time t, in particular r_t).
- Since r_t is a Markov process, $P(t,T)$ depends on the history $(r_s)_{s\le t}$ *only through the current value* r_t. So we can write $P(t,T) = p(t, r_t; T)$ for some deterministic function p.
- The **log-affine form** asserts: $\log p(t,r;T) = a(t,T) - b(t,T)\cdot r$, or equivalently, $P(t,T) = e^{a(t,T)-b(t,T)r_t}$. This means the log-price is a simple linear function of the current rate. When r_t rises by 1 unit, $\log P$ falls by $b(t,T)$ units – $b(t,T)$ is essentially the **duration** of the bond.

5.2 The Feynman-Kac approach: from expectation to PDE

Rather than computing the expectation directly (which would require knowing the full distribution of $\int_t^T r_s \, \mathrm{d}s$), we use the **Feynman-Kac theorem** to convert the pricing problem into a *partial differential equation* (PDE).

Feynman-Kac theorem (informal statement): If X_t is a diffusion process and $f(t,x)$ solves a certain PDE, then $f(t,X_t) = \mathbb{E}\left[g(X_T)\exp\left(-\int_t^T c(s,X_s)\,\mathrm{d}s\right) \,\middle|\, X_t = x\right]$.

For our problem, $X_t = r_t$ (the CIR process under $\mathbb{Q}$), $g \equiv 1$ (the payoff is \$1 at T), and $c(s,r) = r$ (the discount rate equals the short rate). The function $p(t,r) := P(t,T)$ then satisfies:

$$\frac{\partial p}{\partial t} + \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r)\frac{\partial p}{\partial r} + \frac{1}{2}\sigma^2 r \frac{\partial^2 p}{\partial r^2} - r\cdot p = 0,$$

with terminal condition $p(T,r) = 1$ for all $r \ge 0$.

Let us read this PDE term by term:

- $\partial_t p$: how fast the price changes due to time passing (time decay).
- $\kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r)\partial_r p$: the drift of r_t under $\mathbb{Q}$ pushes the price.
- $\frac{1}{2}\sigma^2 r\, \partial_{rr} p$: the convexity correction from the stochastic diffusion (the Itô correction term).
- $-r\cdot p$: the discounting at rate r (the "kill" term).

The PDE says: the infinitesimal change in value (time decay + drift effect + convexity) exactly equals the interest accrued. This is the continuous-time analogue of the bond pricing identity $\partial V/\partial t = rV$ in deterministic finance.

5.3 (i) Proving the log-affine representation

The ansatz. We guess that the solution takes the log-affine form:

$$p(t, r; T) = \exp\big(a(\tau) - b(\tau)r\big), \qquad \tau := T - t \geq 0.$$

(We write $a(\tau)$ and $b(\tau)$ rather than $a(t, T)$ and $b(t, T)$ to emphasise that they depend only on the time-to-maturity $\tau = T - t$, not on t and T separately.)

Why is this guess reasonable? In the deterministic case ($\sigma = 0$), the bond price would be $e^{-r\tau}$ (discounting at constant rate r). Our guess generalises this by allowing the "effective rate" to be $b(\tau)r$ and adding an intercept $a(\tau)$.

Boundary conditions at $\tau = 0$: $p(T, r; T) = 1 = e^0$, so $a(0) = 0$ and $b(0) = 0$.

Substituting into the PDE. Let $P = e^{a-br}$. Compute each derivative:

$$\begin{aligned}
\frac{\partial P}{\partial t} &= \frac{\partial P}{\partial \tau} \cdot \frac{\partial \tau}{\partial t} = -(a'(\tau) - b'(\tau)r)\, P, \\
\frac{\partial P}{\partial r} &= -b(\tau)\, P, \\
\frac{\partial^2 P}{\partial r^2} &= b(\tau)^2\, P.
\end{aligned}$$

Substituting into the PDE and dividing by $P > 0$:

$$-(a' - b'r) + \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r)(-b) + \tfrac{1}{2}\sigma^2 r \cdot b^2 - r = 0.$$

Expanding:

$$-a' + b'r - \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} b + \kappa_{\mathbb{Q}} r b + \tfrac{1}{2}\sigma^2 b^2 r - r = 0.$$

Grouping by powers of r:

$$\begin{aligned}
\text{constant in } r: &\quad -a' - \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} b = 0, \\
\text{linear in } r: &\quad b' + \kappa_{\mathbb{Q}} b + \tfrac{1}{2}\sigma^2 b^2 - 1 = 0.
\end{aligned}$$

The ODE system. We have obtained two ODEs:

$$b'(\tau) = 1 - \kappa_{\mathbb{Q}} b(\tau) - \tfrac{1}{2}\sigma^2 b(\tau)^2, \qquad b(0) = 0, \tag{1}$$

$$a'(\tau) = \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}}\, b(\tau), \qquad a(0) = 0. \tag{2}$$

Equation (1) is a **Riccati ODE** (a first-order nonlinear ODE of the form $y' = p + qy + ry^2$). Equation (2) is just an integral once b is known.

Solving the Riccati ODE. Define $\gamma := \sqrt{\kappa_{\mathbb{Q}}^2 + 2\sigma^2}$. Note $\gamma > \kappa_{\mathbb{Q}} > 0$.

The Riccati ODE (1) can be solved by substitution. The exact solution is:

$$b(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma + \kappa_{\mathbb{Q}})(e^{\gamma\tau} - 1) + 2\gamma}.$$

Let us verify the boundary condition: at $\tau = 0$, numerator is 0 and denominator is $2\gamma > 0$, so $b(0) = 0$. ✓

For small τ: using $e^{\gamma\tau} \approx 1 + \gamma\tau$, $b(\tau) \approx \frac{2\gamma\tau}{2\gamma} = \tau$. So at short maturities, $b \approx \tau$ and $P(t, T) \approx e^{-r_t\tau}$ – exactly the deterministic discounting formula. The CIR formula reduces to deterministic discounting at short maturities.

Solving the a ODE. Substituting and integrating:

$$a(\tau) = \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} \int_0^\tau b(u)\ \mathrm{d}u = \frac{2\kappa_{\mathbb{Q}}\theta_{\mathbb{Q}}}{\sigma^2} \log\left(\frac{2\gamma e^{(\kappa_{\mathbb{Q}}+\gamma)\tau/2}}{(\gamma+\kappa_{\mathbb{Q}})(e^{\gamma\tau}-1)+2\gamma}\right).$$

The result.

$$\boxed{\log P(t,T) = a(T-t) - b(T-t)\, r_t,}$$

with b and a as above. The zero-coupon yield is therefore:

$$y(t,T) = -\frac{\log P(t,T)}{T-t} = \frac{b(T-t)}{T-t}\, r_t - \frac{a(T-t)}{T-t}.$$

This is a linear function of r_t – the entire yield curve moves up and down proportionally to r_t.

Intuition

The function $b(T-t)$ plays the role of **duration** in the CIR model. For short maturities, $b(\tau) \approx \tau$ – the price is approximately the deterministic discount factor. As $\tau \to \infty$, $b(\tau) \to \frac{2}{\gamma+\kappa_{\mathbb{Q}}}$, a finite constant. This means that very long bonds have bounded sensitivity to r_t, unlike in a Vasicek model where $b(\tau) \to \infty$. This is the mean reversion in action: the CIR model "caps" the interest rate sensitivity of very long bonds.

Worked Example

With $\kappa_{\mathbb{Q}} = 0.35$, $\sigma = 0.05$, $\theta_{\mathbb{Q}} = 0.0343$:

$$\gamma = \sqrt{0.35^2 + 2\times 0.0025} = \sqrt{0.1225 + 0.005} = \sqrt{0.1275} \approx 0.357.$$

For $\tau = 5$ years and $r_0 = 0.02$:

$$b(5) = \frac{2(e^{0.357\times 5}-1)}{(0.357+0.35)(e^{1.785}-1)+2\times 0.357} = \frac{2(5.96-1)}{0.707\times 4.96 + 0.714} \approx \frac{9.92}{4.22} \approx 2.35.$$

So $\log P(0,5) = a(5) - 2.35\times 0.02 \approx a(5) - 0.047$. The five-year yield is $y(0,5) = 2.35/5 \times r_0 - a(5)/5$.

5.4 (ii) SDE satisfied by $P(t,T)$

Now we use Itô's lemma on $P(t,T) = e^{a(\tau)-b(\tau)r_t}$ where $\tau = T-t$ is decreasing in t.

Under $\mathbb{Q}$, r_t satisfies:

$$\mathrm{d}r_t = \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\ \mathrm{d}t + \sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{Q}}.$$

Applying Itô's formula to $f(t,r_t) = e^{a(T-t)-b(T-t)r_t}$:

$$\mathrm{d}P = \left(\partial_t P + \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\partial_r P + \tfrac{1}{2}\sigma^2 r_t \partial_{rr} P\right)\ \mathrm{d}t + \partial_r P \cdot \sigma\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{Q}}.$$

The PDE tells us that the $\mathrm{d}t$ bracket equals $r_t P$. And $\partial_r P/P = -b(t,T)$. Hence:

$$\frac{\mathrm{d}P(t,T)}{P(t,T)} = r_t\ \mathrm{d}t - \sigma b(t,T)\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{Q}}.$$

Or equivalently:

$$\mathrm{d}P(t,T) = P(t,T)\big[r_t\ \mathrm{d}t - \sigma b(t,T)\sqrt{r_t}\ \mathrm{d}W_t^{\mathbb{Q}}\big].$$

Reading this equation:

- The drift of the bond under $\mathbb{Q}$ is $r_t\ \mathrm{d}t$ – the bond earns exactly the risk-free rate. This is precisely the martingale condition: after discounting by B_t, no systematic gain remains.
- The volatility of the bond is $\sigma b(t,T)\sqrt{r_t}$. This is:
  - proportional to $b(t,T)$: longer-maturity bonds (larger b) are more volatile,
  - proportional to $\sqrt{r_t}$: bond volatility is higher when rates are higher,
  - proportional to σ: the parameter σ controls how much bond prices fluctuate.

Open-ended questions

- As $T \to \infty$, what does the yield $y(0,T)$ converge to in the CIR model? Is this a reasonable long-run yield level?
- If we observed two bonds of different maturities, could we identify $\kappa_{\mathbb{Q}}$, $\theta_{\mathbb{Q}}$, and σ from their prices alone? How many bonds do you need?

6 Question 7: Fitting the initial yield curve

6.1 The problem: three parameters are not enough

We have a model $(\kappa_{\mathbb{Q}}, \theta_{\mathbb{Q}}, \sigma)$ with three parameters. Once we fix r_0 (today's short rate), the model completely determines the entire yield curve $T \mapsto y^{model}(0,T)$ via the CIR formula. But the real market gives us a yield $y^{market}(0,T)$ for every maturity T, and in general:

$$y^{model}(0,T) \neq y^{market}(0,T) \qquad \text{for most } T.$$

This is a serious problem. If the model misprices the initial yield curve, it will also misprice derivatives (since their prices are functions of bond prices). A **term structure model that cannot even match today's market** is not fit for use in derivatives pricing.

Intuition

Think of it this way. The yield curve on any given day has a complex, irregular shape – perhaps humped at the 5-year point, with a kink at 30 years due to pension fund demand. A three-parameter family of curves (CIR) cannot match all these features exactly. It is like trying to fit an arbitrary polynomial with only a quadratic – you can get the general shape, but not every detail.

6.2 The remedy: time-dependent $\theta_t^{\mathbb{Q}}$

The standard fix (Hull and White, 1990; applied to CIR by Rogers, 1995) is to replace the constant long-run mean $\theta_{\mathbb{Q}}$ with a **deterministic function of time** $\theta_t^{\mathbb{Q}}$. The modified model is:

$$\mathrm{d}r_t = \kappa_{\mathbb{Q}}\big(\theta_t^{\mathbb{Q}} - r_t\big)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad r_0 > 0.$$

Now $\theta_t^{\mathbb{Q}}$ is an entire function, giving us infinitely many degrees of freedom. We can choose it to match the initial yield curve exactly.

What is preserved?

- The diffusion term $\sigma\sqrt{r_t}$ is unchanged.
- The affine structure is preserved: bond prices are still exponential-affine in r_t, so we retain tractability.
- The Feller positivity condition still holds locally (at each t, the local drift is $\kappa_{\mathbb{Q}}\theta_t^{\mathbb{Q}} > 0$ away from zero).

What changes? The model is no longer time-homogeneous. The coefficients $a(t,T)$ and $b(t,T)$ in the log-affine formula now depend on the whole path of $\theta_u^{\mathbb{Q}}$ for $u \in [t,T]$, not just on $\tau = T - t$.

6.3 Bond price formula with time-dependent $\theta_t^{\mathbb{Q}}$

The bond pricing PDE becomes:

$$\partial_t p + \kappa_{\mathbb{Q}}(\theta_t^{\mathbb{Q}} - r)\partial_r p + \tfrac{1}{2}\sigma^2 r \partial_{rr} p - rp = 0, \quad p(T,r) = 1.$$

The ansatz $p = e^{A(t,T) - B(t,T)r}$ still works. The B equation is the same Riccati ODE (independent of $\theta_t^{\mathbb{Q}}$):

$$-\partial_t B = 1 - \kappa_{\mathbb{Q}} B - \tfrac{1}{2}\sigma^2 B^2, \quad B(T,T) = 0.$$

This can be rewritten in $\tau = T - t$: $b'(\tau) = 1 - \kappa_{\mathbb{Q}} b - \frac{1}{2}\sigma^2 b^2$ with $b(0) = 0$ – the same as before. So $b(\tau)$ is unchanged.

The A equation becomes:

$$-\partial_t A = \kappa_{\mathbb{Q}} \theta_t^{\mathbb{Q}} B(t,T), \quad A(T,T) = 0,$$

which integrates to:

$$A(0,T) = \kappa_{\mathbb{Q}} \int_0^T \theta_u^{\mathbb{Q}}\, b(T-u)\, \mathrm{d}u.$$

6.4 Calibrating $\theta_t^{\mathbb{Q}}$ to the initial curve

At $t = 0$, we observe the market yield curve $T \mapsto y^{mkt}(0,T)$, which gives the market discount curve:

$$P^{mkt}(0,T) = e^{-T\, y^{mkt}(0,T)}.$$

The model must match: $P^{model}(0,T) = P^{mkt}(0,T)$ for all T.

Since $P^{model}(0,T) = \exp(A(0,T) - b(T)r_0)$ where we write $b(T) := B(0,T)$ (the Riccati solution evaluated at $\tau = T$), matching gives:

$$A(0,T) = \log P^{mkt}(0,T) + b(T)r_0.$$

Substituting the formula for A:

$$\kappa_{\mathbb{Q}} \int_0^T \theta_u^{\mathbb{Q}}\, b(T-u)\, \mathrm{d}u = \log P^{mkt}(0,T) + b(T)r_0.$$

Define $\Psi(T) := \log P^{mkt}(0,T) + b(T)r_0$. This is a known function of T (computable from market data and the Riccati solution). Then $\theta^{\mathbb{Q}}$ must satisfy:

$$\boxed{\Psi(T) = \kappa_{\mathbb{Q}} \int_0^T \theta_u^{\mathbb{Q}}\, b(T-u)\, \mathrm{d}u.}$$

This is a **Volterra integral equation of the first kind**: it asks "which function $\theta_u^{\mathbb{Q}}$ produces a given convolution with b?"

Numerical recipe. On a maturity grid $0 = T_0 < T_1 < \cdots < T_n$:

1. From market data, compute $P^{mkt}(0,T_i)$ and hence $\Psi(T_i)$.
2. Solve the Riccati ODE for $b(T)$ on the grid (numerically if needed).
3. Approximate the convolution integral using piecewise-constant $\theta_u^{\mathbb{Q}} = \theta_{T_{k-1}}^{\mathbb{Q}}$ for $u \in [T_{k-1}, T_k)$.
4. This turns the integral equation into a lower-triangular linear system, solvable sequentially: given θ values at earlier dates, solve for $\theta_{T_k}^{\mathbb{Q}}$ at each new maturity.

Worked Example

Illustrative two-point example. Suppose $r_0 = 0.02$, $\kappa_{\mathbb{Q}} = 0.35$, $\sigma = 0.05$, and:

$$P^{mkt}(0,1) = 0.980, \quad P^{mkt}(0,2) = 0.955.$$

From the Riccati solution: $b(1) \approx 0.92$, $b(2) \approx 1.70$.
$\Psi(1) = \log(0.980) + 0.92 \times 0.02 \approx -0.0202 + 0.0184 = -0.0018$.
For the first interval, with piecewise-constant $\theta_u^{\mathbb{Q}} = \theta_1$:

$$\kappa_{\mathbb{Q}}\theta_1 \int_0^1 b(1-u)\,\mathrm{d}u \approx 0.35\theta_1 \times 0.45 \approx 0.158\theta_1 = -0.0018,$$

so $\theta_1 \approx -0.011$. (This negative value just illustrates the sensitivity; with realistic market data the values are positive.)

6.5 How do we get κ and σ?

The time-dependent $\theta_t^{\mathbb{Q}}$ can absorb the initial curve for any fixed $(\kappa_{\mathbb{Q}}, \sigma)$. So how do we choose $\kappa_{\mathbb{Q}}$ and σ?

Route 1: Historical estimation. Use daily or weekly short-rate data (e.g., 3-month T-bill yields as a proxy for r_t) to estimate the $\mathbb{P}$ parameters $(\kappa_{\mathbb{P}}, \theta_{\mathbb{P}}, \sigma)$. Since σ is the same under $\mathbb{P}$ and $\mathbb{Q}$, this gives σ directly. For $\kappa_{\mathbb{Q}}$, you also need the market price of risk coefficient η, which requires cross-sectional bond data.

Route 2: Calibration to derivative prices. Fix $(\kappa_{\mathbb{Q}}, \sigma)$ and minimise the pricing error across a panel of liquid cap/floor or swaption prices. Since these instruments are sensitive to σ and $\kappa_{\mathbb{Q}}$ in distinct ways, the two parameters can in principle be identified.

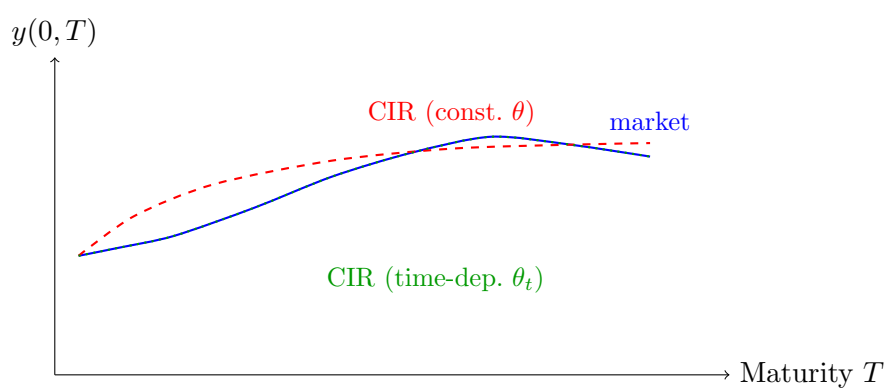

Figure 3: Motivation for time-dependent $\theta_t^{\mathbb{Q}}$. The three-parameter CIR model (dashed red) cannot replicate the hump in the market yield curve. The time-dependent extension (dotted green) fits exactly by construction.

Open-ended questions

- If $\theta_t^{\mathbb{Q}}$ is calibrated daily to market data, what does the time series of $\theta_t^{\mathbb{Q}}$ tell us about the market's risk-neutral expectations?

- Should we smooth $\theta_t^{\mathbb{Q}}$ (e.g., constrain it to be a polynomial in t)? What is the trade-off?

7 Question 8: Interest rate swaps

7.1 (i) What is an interest rate swap?

The practical motivation. Imagine a company, call it ABC Corp, that has issued a \$100 million floating-rate loan. The interest payments reset every quarter and are linked to the 3-month LIBOR rate. ABC Corp is exposed to interest rate risk: if LIBOR rises from 3% to 6%, their quarterly payments double. The CFO wants to eliminate this uncertainty and pay a fixed amount each quarter instead.

On the other side, a bank (XYZ Bank) has a fixed-rate mortgage portfolio paying 5% fixed. The bank funds itself with short-term deposits paying a floating rate. If rates rise, the bank's funding cost rises but its income is fixed – a bad position. The bank wants to *receive* fixed and *pay* floating.

These two parties can enter an **interest rate swap (IRS)**: ABC pays a fixed rate to XYZ and receives LIBOR, while XYZ pays LIBOR to ABC and receives fixed. After netting, ABC's effective obligation is fixed (it pays fixed to XYZ and receives floating from XYZ which offsets its loan), while XYZ has transformed its fixed income into floating income.

The formal definition. An **interest rate swap** is an OTC derivative contract between two counterparties (call them the *fixed-rate payer* and the *floating-rate payer*) to exchange cash flows on specified dates:

- The **fixed leg**: a fixed interest rate K applied to a notional N over accrual periods $[T_{i-1}, T_i]$ for $i = a+1, \ldots, b$.
- The **floating leg**: a floating rate (in our model, set by the short rate) applied to N over the same periods.

The payment dates are $T_{a+1}, \ldots, T_b$, the swap starts at T_a (the *start date*), and we enter the swap at time $t \leq T_a$. We write $\delta_i = T_i - T_{i-1}$ for the day-count fraction (accrual period length).

Cash flows. For a **payer swap** (you pay fixed, receive floating):

- At each date T_i: you **pay** $NK\delta_i$ (fixed).
- At each date T_i: you **receive** $N\big(e^{\int_{T_{i-1}}^{T_i} r_s \, \mathrm{d}s} - 1\big)$ (floating, approximately $Nr_{T_{i-1}}\delta_i$ in discrete terms).

The floating leg in terms of bond prices. In the single-curve framework with continuous compounding, the floating leg payment at T_i (set at T_{i-1}) can be valued as:

$$N\big(P(t, T_{i-1}) - P(t, T_i)\big)$$

because a floating payment at T_i on notional N over $[T_{i-1}, T_i]$ equals a bond maturing at T_{i-1} minus a bond maturing at T_i (a telescoping argument).

Summing over all periods, the total floating leg PV is:

$$PV_t^{float} = N\big(P(t, T_a) - P(t, T_b)\big).$$

Intuition

This telescoping is elegant and worth understanding. The floating leg is equivalent to: receive \$1 at T_a and pay \$1 at T_b. If you lend \$1 at T_a at the prevailing floating rate, you receive the floating coupons and get \$1 back at T_b. The PV of "receive \$1 at T_a, pay \$1 at T_b" is exactly $P(t, T_a) - P(t, T_b)$. The floating coupon payments in between are the "interest" you earn, and they net out to this difference.

The **fixed leg** PV is straightforward:

$$PV_t^{fix} = NK \sum_{i=a+1}^{b} \delta_i P(t, T_i).$$

7.2 (ii) The swap rate

Definition. The **par swap rate** (or simply *swap rate*) at time t, $S(t; T_a, T_b)$, is the fixed rate K that makes the payer swap worth zero at inception:

$$V_t^{payer} = N\big(P(t, T_a) - P(t, T_b)\big) - NK \sum_{i=a+1}^{b} \delta_i P(t, T_i) = 0.$$

Solving for K:

$$\boxed{S(t; T_a, T_b) = \frac{P(t, T_a) - P(t, T_b)}{\sum_{i=a+1}^{b} \delta_i \, P(t, T_i)}.}$$

Economic interpretation.

- **Numerator** $P(t, T_a) - P(t, T_b)$: this is the PV of the floating leg. It is the value of receiving \$1 at T_a and paying \$1 at T_b – the "spread" across the swap's life.
- **Denominator** $\sum_{i=a+1}^{b} \delta_i P(t, T_i)$: this is called the **swap annuity** or **DV01** (dollar value of 1 basis point, scaled). It is the PV of receiving 1 unit of currency per period over the swap's life. It converts the ratio into an annualised rate.
- The swap rate is therefore: "PV of floating leg divided by the annuity". In words: S is the fixed rate that makes the fixed-leg PV equal the floating-leg PV.

Worked Example

Suppose $t = 0$, $T_a = 0$, $T_b = 2$ years, annual payments ($\delta_1 = \delta_2 = 1$), and $P(0, 1) = 0.95$, $P(0, 2) = 0.88$.

$$S(0; 0, 2) = \frac{P(0, 0) - P(0, 2)}{P(0, 1) + P(0, 2)} = \frac{1 - 0.88}{0.95 + 0.88} = \frac{0.12}{1.83} \approx 6.56\%.$$

So the 2-year swap rate is 6.56%. This means: if you enter a 2-year payer swap today paying 6.56% fixed annually, the present value of the fixed payments exactly equals the present value of the floating payments. The swap costs nothing to enter (ignoring bid-ask spreads and counterparty risk).

Swap rate as a function of ZCB prices. The formula above makes this explicit: $S(t; T_a, T_b)$ is entirely determined by the zero-coupon bond prices $P(t, T_a), P(t, T_{a+1}), \ldots, P(t, T_b)$ at time t. No additional information is needed. In the CIR model, each $P(t, T_i) = e^{a(T_i - t) - b(T_i - t) r_t}$, so the swap rate is a deterministic function of r_t alone.

Open-ended questions

- In reality, floating legs are linked to LIBOR/SOFR, not to the theoretical short rate. How does this change the valuation? (This is the "multi-curve" problem.)
- Is the swap rate $S(t; T_a, T_b)$ a martingale under some measure? If so, under which one?

8 Question 9: Swaptions

8.1 (i) What is a swaption?

Financial motivation. Continuing our story from Question 8: ABC Corp wants to enter a swap in 2 years to hedge floating-rate exposure, but is not sure whether it will actually need the hedge (perhaps its floating-rate debt may be refinanced early). ABC wants the *right* – but not the obligation – to enter the swap. This is a **swaption**.

More concretely:

- A **payer swaption** with expiry T_a and strike K gives the holder the right to *enter* a payer swap (pay fixed K, receive floating) starting at T_a.
- The holder exercises if and only if the market swap rate $S(T_a; T_a, T_b) > K$: if rates have risen, the option to pay the lower fixed rate K is valuable.

Swaptions are one of the most liquid interest rate derivative instruments. They are used to hedge the optionality embedded in mortgages (prepayment options) and callable bonds.

Formal definition. A **European payer swaption** with notional N, expiry T_a, swap tenor $[T_a, T_b]$, payment dates $T_{a+1}, \ldots, T_b$, and strike K has payoff at T_a:

$$\Pi_{T_a}^{pay} = N\left(V_{T_a}^{payer}\right)^+ = N\left(P(T_a, T_a) - P(T_a, T_b) - K\sum_{i=a+1}^{b} \delta_i P(T_a, T_i)\right)^+.$$

Since $P(T_a, T_a) = 1$:

$$\Pi_{T_a}^{pay} = N\left(1 - P(T_a, T_b) - K\sum_{i=a+1}^{b} \delta_i P(T_a, T_i)\right)^+.$$

Using the annuity $A_{T_a} = \sum_{i=a+1}^{b} \delta_i P(T_a, T_i)$ and the swap rate formula:

$$\Pi_{T_a}^{pay} = N \cdot A_{T_a} \cdot (S(T_a; T_a, T_b) - K)^+.$$

This is an option on the swap rate S: a call option (for a payer swaption).

8.2 (ii) Pricing formula

General risk-neutral formula. The time-t price of the payer swaption is:

$$V_t^{pay} = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^{T_a} r_s \, \mathrm{d}s}\, \Pi_{T_a}^{pay}\right] = N \cdot \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^{T_a} r_s \, \mathrm{d}s} \cdot A_{T_a} \cdot (S(T_a; T_a, T_b) - K)^+\right].$$

The swap-annuity measure. Define the **annuity numeraire**:

$$A_t := A(t; T_a, T_b) = \sum_{i=a+1}^{b} \delta_i P(t, T_i).$$

Since A_t is a traded (non-negative) asset, it can serve as a numeraire. Under the corresponding **annuity measure** $\mathbb{Q}^A$:

$$V_t^{pay} = N \cdot A_t \cdot \mathbb{E}_t^{\mathbb{Q}^A}\left[(S(T_a; T_a, T_b) - K)^+\right].$$

The swap rate $S(t; T_a, T_b) = (P(t, T_a) - P(t, T_b))/A_t$ is a $\mathbb{Q}^A$**-martingale** (because the numerator is the difference of two prices and the denominator is the numeraire). This is the theoretical foundation for Black's formula.

Black's swaption formula. If we assume the swap rate is log-normally distributed under $\mathbb{Q}^A$, we obtain the **Black swaption formula**:

$$V_t^{pay} = N \cdot A_t \Big(S_t \, \Phi(d_1) - K \, \Phi(d_2) \Big),$$

where $S_t := S(t; T_a, T_b)$ is today's swap rate and:

$$d_{1,2} = \frac{\ln(S_t/K) \pm \frac{1}{2}\sigma_{swap}^2 (T_a - t)}{\sigma_{swap}\sqrt{T_a - t}},$$

and σ_{swap} is the **swaption (Black) implied volatility**.

Intuition

Black's formula treats the swaption exactly like a call option on a stock (the "stock" being the swap rate S_t), but with the bond annuity A_t playing the role of the discount factor. The formula is the fixed income analogue of the Black-Scholes formula. The market quotes swaptions in terms of σ_{swap} (implied vol), not in terms of dollar prices.

CIR pricing via Jamshidian's decomposition. In the CIR model (or any one-factor short-rate model), we can be more precise. The swaption payoff can be rewritten as a portfolio of bond *options*:

1. Let coupon weights $c_i = K\delta_i$ for $i = a+1, \ldots, b-1$ and $c_b = 1 + K\delta_b$. Then:

$$\Pi_{T_a}^{pay} = N \left(1 - \sum_{i=a+1}^{b} c_i P(T_a, T_i) \right)^+.$$

2. Since $P(T_a, T_i) = e^{a(T_i - T_a) - b(T_i - T_a) r_{T_a}}$ is a *monotone decreasing* function of r_{T_a} (in a one-factor model, all bond prices move together), there exists a unique *critical rate* $r^\star$ such that:

$$\sum_{i=a+1}^{b} c_i P(T_a, T_i; r^\star) = 1.$$

3. **Jamshidian's trick**: the payoff is positive iff $r_{T_a} < r^\star$ (low rates make bond prices high and the swap in-the-money for the payer). Define $K_i = P(T_a, T_i; r^\star)$ as the "strike" for the i-th bond option. Then:

$$\Pi_{T_a}^{pay} = N \sum_{i=a+1}^{b} c_i \Big(K_i - P(T_a, T_i) \Big)^+.$$

   This is a *portfolio of put options on zero-coupon bonds*, with notional Nc_i and strike K_i for the i-th bond.

4. Each zero-coupon bond option can be priced in closed form in the CIR model using the non-central chi-squared distribution (since $P(T_a, T_i)$ is a function of r_{T_a}, whose distribution is known exactly).

The CIR payer swaption price is therefore:

$$V_t^{pay} = N \sum_{i=a+1}^{b} c_i \, \text{ZCBput}(t;\, T_a, T_i, K_i),$$

where $\text{ZCBput}(t;\, T_a, T_i, K_i)$ is the time-t price of a put option on the T_i-maturity ZCB, with expiry T_a and strike K_i, priced using the CIR distribution.

Worked Example

What Jamshidian's trick achieves. Pricing an option on a coupon bond directly requires knowing the joint distribution of $P(T_a, T_{a+1}), \ldots, P(T_a, T_b)$. In a multi-factor model, this is complicated. In a one-factor model like CIR, all these bond prices are determined by the single state variable r_{T_a}, so the joint distribution reduces to the univariate distribution of r_{T_a} – which we know exactly (non-central chi-squared). Jamshidian's decomposition is only possible in a one-factor world.

This is one practical limitation of CIR: it cannot price swaptions and caps consistently without implying a perfect correlation between all bond maturities. In reality, swaption skews and cap-floor basis suggest multi-factor dynamics.

Open-ended questions

- Under what conditions does the Black swaption formula overstate or understate the CIR price? When are they approximately equal?
- How would you calibrate the CIR model to a panel of swaptions $V^{pay}(T_a, T_b, K)$ for multiple (T_a, T_b, K)? Is the system over- or under-determined?

References and suggested reading

- Cox, J.C., Ingersoll, J.E., Ross, S.A. (1985): "A Theory of the Term Structure of Interest Rates." *Econometrica*, 53(2), 385–407. *The original CIR paper.*
- Brigo, D. and Mercurio, F. (2006): *Interest Rate Models — Theory and Practice.* Springer. *The definitive reference; see Chapters 3 (CIR), 6 (swaptions), and 12 (term structure fitting).*
- Filipović, D. (2009): *Term-Structure Models.* Springer. *Rigorous mathematical treatment of affine models.*
- Shreve, S. (2004): *Stochastic Calculus for Finance II.* Springer. *Chapter 9 covers interest rate models.*
- Jamshidian, F. (1989): "An exact bond option formula." *Journal of Finance*, 44(1), 205–209. *Original paper for the decomposition in Question 9.*
- Hull, J. and White, A. (1990): "Pricing interest rate derivative securities." *Review of Financial Studies*, 3(4), 573–592. *The source for time-dependent drift extension.*
- Yamada, T. and Watanabe, S. (1971): "On the uniqueness of solutions of stochastic differential equations." *J. Math. Kyoto Univ.*, 11, 155–167. *For the strong solution result in Question 3.*

Simulation and coding roadmap (iteration 2)

When we move to the coding phase, the following tasks are planned:

1. **CIR simulation under $\mathbb{P}$:** Euler-Maruyama and Milstein schemes for the square-root diffusion. Plot sample paths, histogram of r_T vs. the theoretical non-central chi-squared density.
2. **Yield curve generation:** Given CIR parameters, compute $b(\tau)$ and $a(\tau)$ on a maturity grid and plot the yield and forward curves.
3. **Initial curve fitting:** Load a FRED yield curve for a chosen date. Solve the Volterra equation numerically to find $\theta_t^{\mathbb{Q}}$. Plot observed vs. fitted yield curve.
4. **Swap rate surface:** Compute the swap rate $S(0; 0, T)$ as a function of tenor T.
5. **Swaption pricing:** Implement Jamshidian's decomposition and plot the swaption price as a function of strike and expiry. Compare with Black's formula.